COMPLETE ACCOUNT · SIX PAPERS UNIFIED

A Circle-Free Theory of π

Six independent definitions of π, the trap-free bridges that prove them one number, the resolution of a normalization interlock, and efficient ways to compute each — assembled into a single self-contained account.

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Status: verified symbolically & numerically

HOW TO READ THIS DOCUMENT

Plain Written for any curious reader, no background assumed. Read only these and you still get the whole story.

Rigorous The same claim for a technical reader: exact conditions, the proof idea, the failure mode.

Variables Every symbol defined in plain words the first time its section uses it. No symbol silently changes meaning.

Worked example A concrete number pushed through the formula, so the abstraction has something to hold onto.

Correction What an earlier draft got wrong and what replaced it. ~~Struck red~~ = old; **green** = fix.

ABSTRACT

One number, wearing six costumes

The constant π appears across mathematics in places that have nothing to do with circles: the growth rate of factorials, the density of square-free whole numbers, the count of grid points in a round region, the volume of a cone, the chance two random integers share no common factor. This paper gives six such definitions — each from a different branch of

mathematics, each stated without circles, angles, trigonometric functions, or polar coordinates — and proves all six define the same number through a network of bridges in which no step secretly smuggles a circle back in.

The keystone is a characterization of the Wallis integral by three circle-free properties (a recurrence, log-convexity, and an asymptotic normalization), in close analogy with the Bohr–Mollerup theorem for the Gamma function. It fixes one exact value, $W(1/2) = \pi_0/4$, from which the Gaussian integral, $\Gamma(1/2)$, the lattice-point density, and the Basel sum $\zeta(2) = \pi_0^2/6$ all follow without circular geometry. We give the full dependency graph, prove it has no cycles, document a circularity an earlier version contained and exactly how it was removed, and close with efficient (and still circle-free) ways to evaluate each definition, plus an honest account of where the work is useful.

PART I · THE RULE OF THE GAME

What "circle-free" forbids, and why

Everyone knows π as "circumference ÷ diameter." The puzzle this paper answers is whether the *other* appearances of π — in counting, in number theory, in volumes — can be proven equal to that one *without* ever leaning on the circle as a proof step. To make that precise we ban three moves, which we call **traps**, because each silently assumes circular geometry even when no circle is named.

VARIABLES

π_0	the common value we are trying to pin down (it will turn out to be 3.14159…). Subscript 0 is a reminder that, until the bridges are built, we may *not* assume the six definitions are equal.
`cos, sin`	the coordinates of a point moving around the unit circle. Using them is using a circle.
`arctan, arcsin`	angle-valued inverse functions; their very definition is "the angle on a circle whose…"

PLAIN Imagine π as one person who keeps turning up at different parties in different costumes — a geometry party, a counting party, a number-theory party. We want to prove it is the same person each time *without* asking them to remove the costume and show the circle underneath. The three forbidden moves below are the three ways one usually "peeks."

TRAP	THE FORBIDDEN MOVE	WHY IT HIDES A CIRCLE
1 · Polar	`(x,y) → (r cosθ, r sinθ)`	cos and sin *are* a point circling the origin.
2 · Power series	`1 − 1/3 + 1/5 − … = π/4`	that is the arctangent series — an angle on a circle.
3 · Integration	`∫ 1/√(1−t²) dt "= arcsin"`	arcsin, and the polar trick for $\int e^{-x^2}$, end at $\Gamma(1/2)=\sqrt{\pi}$ through circular machinery.

RIGOROUS The traps constrain proof *strategy*, not truth. A statement can be true and provable by a trapped method while also admitting a trap-free proof; the taxonomy marks where circularity *enters* an argument, not where it is logically *necessary*. Part VII shows, via periods and motives, that all six definitions compute the period of one object — so the circle is never logically eliminable, only avoidable as an explicit step.

PART II · THE CAST

Six circle-free definitions of π

Each definition below is self-contained and uses only the tools of its own field. None mentions a circle. The theorem of this paper is that all six produce the identical number π_0.

DEFINITION I

COMBINATORICS

DEFINITION II

SOLID GEOMETRY

Factorial constant

$$\pi_0 = \lim 16^n (n!)^4 / (n\cdot((2n)!)^2)$$

Volumetric ratio

$$\kappa = abc / (2r^2h), \text{ at equal volume}$$

DEFINITION III

GEOMETRY OF NUMBERS

Lattice-point density

$$\lim \#\{a^2+b^2 \le N^2\} / N^2$$

DEFINITION IV

ARITHMETIC

Square-free density

$$6/\pi^2 = \prod_p (1 - 1/p^2)$$

DEFINITION V

ANALYSIS

Wallis integral

$$\pi_0 = 4\int_0^1 \sqrt{(1-t^2)}\, dt$$

DEFINITION VI

PROBABILITY

Coprime probability

$$P(\gcd(a,b)=1) = 6/\pi^2$$

I — The factorial constant

VARIABLES

- `n!` “n factorial,” the product $1\cdot2\cdot3\cdots n$. Example: $4! = 24$.
- `lim` the value a sequence settles onto as n grows without bound.
- a_n the n-th term of the recurrence below; the sequence whose limit is π_0.

PLAIN Factorials count arrangements: $4! = 24$ is the number of ways to line up four books.

Take a particular ratio of these counting numbers, and as n grows the ratio settles onto π — with nothing but multiplication and division in sight.

RIGOROUS The limit is computed by the recurrence $a_1 = 4$, $a_{n+1} = a_n \cdot 4n(n+1)/(2n+1)^2$. Since $4n(n+1)/(2n+1)^2 = 1 - 1/(2n+1)^2 < 1$, the sequence strictly decreases and stays above 0, so by monotone convergence it has a limit — that limit is π_0. Equivalently, with the central binomial coefficient $A_n = (2n \text{ choose } n)/4^n \sim 1/\sqrt{\pi n}$, squaring and rearranging gives the same constant. No geometry enters.

WORKED EXAMPLE The first terms: $a_1 = 4$, $a_2 = 4\cdot(4\cdot1\cdot2)/3^2 = 32/9 = 3.555\ldots$, $a_3 = 3.555\ldots\cdot(4\cdot2\cdot3)/5^2 = 3.413\ldots$, marching down toward 3.14159. At a_{1000} the value is 3.14237; the error shrinks like 1/n.

II — The volumetric ratio

VARIABLES

`r, h`	the base “radius” (half-width of the base region) and height of a cone.
`a, b, c`	three mutually perpendicular edge lengths of a right tetrahedron (a corner pyramid).
`κ (kappa)`	the constant relating the cone-base area to r^2. We prove $\kappa = \pi_0$.

PLAIN One shape (the cone) carries π in its volume; the other (the corner pyramid) does not — its volume is just abc/6, plain algebra. Build the two so they hold the same amount of water, and the ratio of their measured dimensions hands you π. You could literally pour water between two vessels and read π off a ruler.

RIGOROUS A cone of base region $\{x^2+y^2 \le r^2\}$ and height h has volume $V_{cone} = (1/3)\kappa r^2 h$ (from Cavalieri’s principle and $\int z^2\,dz = z^3/3$, polynomial calculus). A right tetrahedron has $V_{tet} = abc/6$ (a determinant; no π). Setting $V_{cone} = V_{tet}$ gives $\kappa = abc/(2r^2h)$. The cone supplies the π-carrying metric; the tetrahedron supplies a π-free reference frame. A useful restatement: multiplying $V_{cone} = (1/3)\pi r^2 h$ by $3/\pi$ gives $3V/\pi = r^2h$, a whole number when r,h are integers — the bridge from geometry to arithmetic.

WORKED EXAMPLE Let a cone have $r = 3$, $h = 4$, so its base “divisor” $r^2h = 36$. A right tetrahedron with $a = 6$, $b = 6$, $c = 6$ has $abc = 216$. Then $\kappa = 216/(2\cdot9\cdot4) = 216/72 = 3.0$ — not yet π, because these particular shapes are not at equal volume. Tuning the dimensions until $V_{cone} = V_{tet}$ drives κ to 3.14159. (The exact tuning is what Part IV computes.)

III — The lattice-point density

VARIABLES

$\mathbb{Z}^2$	the grid of all points (a,b) with whole-number coordinates.
#{…}	“the number of points satisfying…”
N	the size of the region; we let it grow without bound.

PLAIN Lay graph paper over a large round region, count the corners that fall inside, and divide by N^2. The bigger the region, the closer that count comes to π. This is π born from pure counting on a grid.

RIGOROUS $\#\{(a,b)\in\mathbb{Z}^2 : a^2+b^2 \le N^2\} = \pi_0 N^2 + E(N)$, where the boundary error $E(N)$ is $O(N)$ (the Gauss circle problem). Dividing by N^2 and taking $N\to\infty$ kills the error term and leaves π_0. Part V proves this equals $4W(1/2)$ by a Riemann-sum argument, with no circular geometry.

WORKED EXAMPLE At $N = 10$, the points with $a^2+b^2 \le 100$ number 317, and $317/100 = 3.17$. At $N = 100$ the ratio is 3.1417; at $N = 1000$, 3.14155. Slowly, with a faint wobble from the boundary, it converges on π.

IV — The square-free density

VARIABLES

square-free an integer no perfect square > 1 divides. $10 = 2·5$ is square-free; $12 = 4·3$ is not.

$\prod_p$ a product running over all prime numbers $p = 2, 3, 5, 7, \dots$

$\zeta(2)$ the sum $1 + 1/4 + 1/9 + 1/16 + \dots$ of reciprocal squares; equals $\pi^2/6$.

PLAIN Pick a whole number at random. The chance it has no repeated prime factor is about 61% — and that percentage is exactly 6 divided by π^2. π is quietly governing a fact about ordinary counting numbers.

RIGOROUS By unique factorization an integer is square-free iff no p^2 divides it; independence across primes gives the Euler product $\prod_p(1-1/p^2) = 1/\zeta(2)$. With $\zeta(2) = \pi_0^2/6$ (proved trap-free in Part VI) the density is $6/\pi_0^2 \approx 0.6079$.

WORKED EXAMPLE Among 1–10, the non-square-free numbers are 4, 8, 9 (and 12,16,18,… later), so 7 of the first 10 are square-free — 70%, noisy this early. Among the first million, the fraction is 0.607927…, hugging $6/\pi^2 = 0.607927\dots$.

V — The Wallis integral

VARIABLES

W(x) the Wallis integral $\int_0^1(1-t^2)^x\,dt$, defined for every $x > -1/2$.

t the integration variable, running from 0 to 1.

PLAIN $W(1/2)$ is the area under the curve $y = \sqrt{(1-t^2)}$ from 0 to 1 — a quarter of a round region's area. Multiply by 4 and you have π. The whole trick of the paper is to pin down this area *without* recognizing the curve as part of a circle.

RIGOROUS $\pi_0 = 4W(1/2)$. The function W obeys the recurrence $W(x) = (2x/(2x+1))W(x-1)$ with $W(0) = 1$ (integration by parts), and at integers $W(n) = \prod_{k=1}^{n} 2k/(2k+1)$ is rational. Part IV pins $W(1/2) = \pi_0/4$ without evaluating the integral by an antiderivative.

WORKED EXAMPLE $W(0) = 1$, $W(1) = \int_0^1(1-t^2)dt = 1 - 1/3 = 2/3$, $W(2) = (4/5)\cdot(2/3) = 8/15$. These are all plain fractions. Only the half-integer value $W(1/2)$ carries π.

VI — The coprime probability

VARIABLES

`gcd(a,b)`	the greatest common divisor of a and b; “coprime” means gcd = 1.
`P(…)`	the long-run probability over random integer pairs.

PLAIN Pick two whole numbers at random; the odds they share no common factor are again about 61% — the same $6/\pi^2$ as the square-free density, now arriving from probability instead of divisibility.

RIGOROUS $P(\gcd(a,b)=1) = \prod_p(1-1/p^2) = 1/\zeta(2) = 6/\pi_0^2$: the probability that no prime divides both a and b, computed prime-by-prime. Definitions IV and VI are two faces of one fact, both governed by $\zeta(2)$.

WORKED EXAMPLE Among the 100 pairs (a,b) with $1 \le a,b \le 10$, exactly 63 are coprime — 63%. The fraction tends to $6/\pi^2 \approx 60.79\%$ as the range grows.

The Wallis integral, and the bridges I $\leftrightarrow$ V $\leftrightarrow$ II

All six definitions connect through one function, $W(x)$. We first record its elementary properties — each trap-free — then bridge it to the factorial constant and to the volume ratio. The remaining gap, one exact value, is closed in Part IV.

VARIABLES

- `W(x)` $\int_0^1(1-t^2)^x\,dt$, the Wallis integral.
- `P(n)` the half-step product $\prod_{k=1}^{n}(2k+1)/(2k+2)$; appears in the telescoping identity.
- `L` the limiting constant $\sqrt{x}\cdot W(x)$ approaches as $x\to\infty$; equals $\sqrt{\pi_0}/2$.

The recurrence and the rational integer values

RIGOROUS Integration by parts on $\int_0^1(1-t^2)^x\,dt$ gives

$$W(x) = \frac{2x}{2x+1}\,W(x-1),\quad W(0) = 1. \tag{1}$$

Iterating from $W(0)=1$ yields, for whole numbers n,

$$W(n) = \prod_{k=1}^{n}\frac{2k}{2k+1} = \frac{2}{3}\cdot\frac{4}{5}\cdot\frac{6}{7}\cdots\frac{2n}{2n+1} \tag{2}$$

Every $W(n)$ is rational — no π, no transcendental ingredient. **Trap-free** (only the product rule and the fundamental theorem of calculus).

Bridge I $\leftrightarrow$ V: the factorial constant

PLAIN The half-integer ladder of W and the factorial formula turn out to be two ways of writing the same shrinking product. Line them up and "n times $W(n)$ squared" homes in on $\pi_0/4$.

RIGOROUS — PROPOSITION 1 Writing $W(n) = 4^n(n!)^2/(2n+1)!$ and squaring,

$$\lim_{n\to\infty} n\cdot W(n)^2 = \pi_0/4. \tag{3}$$

The product $n\cdot 16^n(n!)^4/((2n+1)!)^2$ equals the Definition-I expression times $(n/(2n+1))^2$, which tends to 1/4; hence its limit is $\pi_0\cdot(1/4) = \pi_0/4$. (Stirling's approximation applied to either form yields $\pi_0/4$ directly.)

WORKED EXAMPLE At $n = 10000$, $W(n) \approx 0.0088623$, and $n\cdot W(n)^2 \approx 0.785377$, already close to $\pi/4 = 0.785398$.

Bridge II ↔ V: the volume ratio

RIGOROUS Filling the cone base with thin rectangular strips (a Riemann sum — not a circle) gives base area $\kappa\, r^2 = 4r^2\int_0^1\sqrt{(1-t^2)}\, dt = 4r^2 W(1/2)$. Hence

$$\kappa = 4W(1/2). \tag{4}$$

Definition II thus reduces to the single value $W(1/2)$. The integral is deliberately left unevaluated — finding its antiderivative would spring Trap 3.

PLAIN — THE ONE THING LEFT TO PROVE Both bridges now hinge on one number: the area $W(1/2)$. Proposition 1 says "$n\cdot W(n)^2 \to \pi_0/4$" for whole numbers; we need the same constant to control the *half*-integer value. That is the keystone, proved next.

$$\textbf{Goal:}\quad W(1/2) = \pi_0/4.$$

A Bohr–Mollerup theorem for the Wallis integral

The Gamma function is singled out among all functions by three properties (a recurrence, log-convexity, a normalization) — that is the Bohr–Mollerup theorem. We prove the exact analogue for W, then read off $W(1/2) = \pi_0/4$.

VARIABLES

log-convex a function whose logarithm never bends the wrong way (curves upward or stays straight).

admissible our name for a function with all three keystone properties: the recurrence (1), log-convexity, and $\sqrt{x}\cdot f(x) \to \lambda$.

λ (lambda) the common asymptotic constant in $\sqrt{x}\cdot f(x) \to \lambda$.

Three structural properties

RIGOROUS — LEMMAS 1-3 **(L1) Log-convexity.** By Hölder's inequality, $W(\lambda x_1+(1-\lambda)x_2) \le W(x_1)^{\lambda} W(x_2)^{1-\lambda}$; taking logs gives convexity. (Hölder descends from AM–GM — pure algebra.)

(L2) Recurrence. Equation (1).

(L3) Gaussian sandwich. For $t \in [0,1]$, $x \ge 1$: $e^{-xt^2-xt^4} \le (1-t^2)^x \le e^{-xt^2}$, from the series $\ln(1-t^2) = -t^2 - t^4/2 - \cdots$. **All trap-free.**

PLAIN Log-convexity means the graph of ln W never kinks the wrong way. The Gaussian sandwich pins the integrand between two bell-curve fences we never have to measure — they only act as upper and lower walls. Together with the ladder rule, these walls will lock W in place.

The asymptotic normalization

RIGOROUS — THEOREM 1 Let $L = \lim_{R\to\infty} \int_0^R e^{-u^2}\,du$ (exists: bounded and increasing). Substituting $u = \sqrt{x}\cdot t$ and applying the sandwich (L3),

$$\sqrt{x} \cdot W(x) \to L, \quad \text{and} \quad L^2 = \pi_0/4. \tag{5}$$

Crucially L is *never evaluated*: its existence is monotonicity, and its value is *forced* by $L^2 = \lim n\cdot W(n)^2 = \pi_0/4$ (Proposition 1). The Gaussian integral is here a comparison fence, not an object needing polar coordinates — which is what keeps Theorem 1 out of Trap 1.

The uniqueness lemma

RIGOROUS — LEMMA 4 (NORMALIZATION UNIQUENESS) Call $f > 0$ *admissible* if it satisfies (1), is log-convex, and $\sqrt{x}\cdot f(x)\to\lambda$ for some finite $\lambda>0$. **Two admissible functions with the same λ are identical.** Proof: with $h = f/g$, the shared recurrence gives $h(x) = h(x-1)$ (period 1); the shared asymptotic gives $h(x)\to 1$. A period-1 function tending to 1 equals 1 everywhere, so $f \equiv g$.

PLAIN Two functions that climb the same ladder, never bend the wrong way, and end at the same far-off height must have been the same function all along. There is no wiggle room left — and that is what fixes $W(1/2)$.

The telescoping identity and the exact value

RIGOROUS — THE ALGEBRAIC HEART Run (1) at half-steps: $W(n+1/2) = W(1/2)\cdot P(n)$ with $P(n)=\prod_{k=1}^{n}(2k+1)/(2k+2)$. A telescoping product reveals

$$W(n)\cdot P(n) = \prod_{k=1}^{n} \frac{2k}{2k+1}\cdot\frac{2k+1}{2k+2} = \prod_{k=1}^{n} \frac{k}{k+1} = \frac{1}{n+1} \tag{6}$$

so $P(n) = 1/[(n+1)W(n)]$, hence $W(n+1/2) = W(1/2)/[(n+1)W(n)]$.

RIGOROUS — THEOREM 2 (KEYSTONE) Apply the asymptotic (5) along the half-integer ladder: $\sqrt{(n+1/2)}\cdot W(n+1/2) \to L$. Substituting the identity and $W(n) \sim L/\sqrt{n}$,

$$\mathbf{W(1/2) = L^2 = \pi_0/4.}$$

With (4), $\kappa = 4W(1/2) = \pi_0$. **Definitions I, II, V now coincide.**

WORKED EXAMPLE — THE IDENTITY IS EXACT At $n = 5$: $W(5) = (2/3)(4/5)(6/7)(8/9)(10/11) = 0.36941\ldots$ and $P(5) = (3/4)(5/6)(7/8)(9/10)(11/12) = 0.45117\ldots$. Their product is $0.16666\ldots = 1/6 = 1/(5+1)$, to the last digit. The telescoping leaves no error — it is algebra, not approximation.

WHY TWO LAYERS ARE NEEDED The asymptotic (5) supplies the constant L; the telescoping identity (6) is an exact rational statement. Neither alone fixes $W(1/2)$ — the asymptotic gives only a one-sided bound, the identity alone leaves L unknown. Together they close the value with zero gap. This two-layer structure is what earlier revisions lacked.

PART V · CLOSING THE LOOP

The lattice bridge, and three classical constants for free

Bridge III ↔ V: lattice points are a Riemann sum

VARIABLES

$\lfloor\cdot\rfloor$ the floor function: round down to the nearest whole number.

$O(N)$ "grows no faster than a constant times N" — the size of the boundary error.

RIGOROUS — THEOREM 3 By symmetry, $\#\{a^2+b^2\le N^2\} = 4\sum_{a=0}^{N}\lfloor\sqrt{(N^2-a^2)}\rfloor + O(N) = 4N\cdot\sum_{a=0}^{N}\sqrt{(1-(a/N)^2)}\cdot(1/N) + O(N)$. The sum is a Riemann sum for $\int_0^1\sqrt{(1-t^2)}dt = W(1/2)$. Hence the count $\sim 4N^2W(1/2) = \pi_0N^2$. **Trap-free**: Riemann-sum convergence is standard analysis; $\sqrt{(1-t^2)}$ is integrated as a function, not named as a curve.

PLAIN Counting grid points under the curve $\sqrt{(1-t^2)}$ is the same as adding the heights of many thin columns — exactly how one approximates an area. Refine the grid and the count becomes the area $W(1/2)$; four times that is π. No circle was named; columns were summed.

Three famous constants, each normally needing a trap

RIGOROUS — THE BACKWARD RECURRENCE (THEOREM 4) The recurrence (1) runs *downward* too. At $x = 1/2$: $W(1/2) = (1/2)W(-1/2)$, so

$$\int_0^1 \frac{1}{\sqrt{(1-t^2)}}\,dt = W(-1/2) = 2W(1/2) = \pi_0/2. \tag{7}$$

This is the arcsine integral — evaluated *without* writing arcsin. Trap 3 disarmed.

RIGOROUS — Γ(1/2) AND THE GAUSSIAN INTEGRAL (THEOREMS 5-6) The Beta form $W(x) = \frac{1}{2} B(\frac{1}{2},x+1) = \Gamma(\frac{1}{2})\Gamma(x+1)/(2\Gamma(x+3/2))$ at $x=1/2$ gives $W(1/2)=\Gamma(1/2)^2/4$, so

$$\Gamma(1/2) = \sqrt{\pi_0}, \quad \int_0^\infty e^{-u^2}\,du = L = \sqrt{\pi_0}/2. \tag{8}$$

Both normally require polar coordinates (Trap 1); here they fall out of the keystone. The Beta–Gamma relation is an algebraic Jacobian on the positive quadrant, *not* a polar map.

WORKED EXAMPLE $\Gamma(1/2) = \sqrt{\pi} = 1.772453\ldots$, and the bell-curve area $\int_{-\infty}^{\infty} e^{-u^2} du = \sqrt{\pi} = 1.772453\ldots$ — the most famous $\sqrt{\pi}$ in mathematics, here obtained with no polar trick.

PART VI · THE DEEPEST BRIDGE

$\zeta(2) = \pi_0^2/6$, without a Fourier series in sight

Definitions IV and VI both rest on $\zeta(2)$. Every classical proof of $\zeta(2)=\pi^2/6$ springs a trap (Euler's uses the sine product; Parseval's uses sin and cos). We reach it through the second logarithmic derivative of W.

VARIABLES

`(ln W)″(0)`	the second derivative of ln $W(x)$, evaluated at $x = 0$: a measure of how sharply that curve bends at the start.
`C`	shorthand for that number, $(\ln W)''(0)$. It will turn out $C = 4 - 2\zeta(2)$.
`ψ′(s)`	the trigamma function, here just notation for the series $\sum_{n\ge 0} 1/(n+s)^2$.
`O`	the odd-square sum $\sum_{k\ge 0} 1/(2k+1)^2 = \pi_0^2/8$.

One clean number, computed two ways

RIGOROUS Define $C = (\ln W)''(0) = W''(0) - W'(0)^2$. Both pieces come from expanding the logarithm — pure polynomial calculus:

$$W'(0) = \int_0^1 \ln(1-t^2)\,dt = 2\ln 2 - 2, \tag{9}$$

$$W''(0) = \int_0^1 [\ln(1-t^2)]^2\,dt = \sum_{k,j\geq 1} 1/[kj(2k+2j+1)]. \tag{10}$$

So C is a definite number from convergent rational series — no trigonometry, and crucially *no value of ζ(2) put in by hand*.

The closure

RIGOROUS — THEOREM 7 The second log-derivative of W equals $\psi'(1) - \psi'(3/2)$. Now $\psi'(1) = \zeta(2)$, and splitting even/odd terms gives $\psi'(3/2) = 4(\tfrac{3}{4}\zeta(2)-1)$. Therefore

$$C = \zeta(2) - 4(\tfrac{3}{4}\zeta(2)-1) = 4 - 2\zeta(2), \quad \text{so} \quad \zeta(2) = (4 - C)/2. \tag{11}$$

Since C is fixed trap-free by (9)–(10), $\zeta(2)$ is an *output*: numerically $C = 0.710132\ldots$ gives $\zeta(2) = 1.644934\ldots = \pi_0^2/6$.

PLAIN C measures how sharply the curve $\ln W$ bends right at the start. Compute it two ways — straight from the area integral (just fractions), and through the standard bending formula — set them equal, and $\zeta(2)$ drops out. We never feed in "$= \pi^2/6$"; it comes out the far end.

WORKED EXAMPLE $\int_0^1 \ln(1-t^2)\,dt = 2\ln 2 - 2 = -0.613706\ldots$ (elementary). Feeding the two integrals in: $C = 0.710132$, so $(4-C)/2 = 1.644934 = \pi^2/6$. The square-free density $6/\pi^2 = 0.607927$ follows immediately — closing Definitions IV and VI.

PART VII · HONESTY ABOUT PROCESS

An error we made, and exactly how we fixed it

An earlier revision of the ζ(2) bridge contained a real circularity, caught in adversarial review. Recording it (and its repair) is part of the work.

THE HOLE

The earlier draft proved $\zeta(2) = (4-C)/2$ cleanly, then converted it to $\pi_0^2/6$ using the auxiliary identity ~~$\Theta = 2W(1/2)^2$~~. That identity is true, but its only classical proof evaluates $\int\sqrt{(1-t^2)}dt$ by the substitution $t = \sin\theta$ — exactly Trap 3. The bridge had quietly smuggled a circle back in.

PLAIN We had two clean halves and tried to glue them with a step that, looked at closely, used the very circle we had banned. A careful reviewer caught it. The fix was not to patch the bad glue but to remove the need for it.

CORRECTION **Dissolve, don't patch.** Both facts the bridge needs are read off one trap-free closed form,

$$W(x) = (\Gamma(1/2)/2)\cdot\Gamma(x+1)/\Gamma(x+3/2), \quad \Gamma(1/2) = \sqrt{\pi_0}.$$

The value $W(1/2)=\pi_0/4$ is established *first* (Theorem 2, by telescoping — no Γ, no trig); the closed form is then a consequence, and C is read off the *same* validated form. The identity ~~$\Theta = 2W(1/2)^2$~~ is never used.

RIGOROUS — THE RESIDUAL SUBTLETY, FLAGGED The repair uses only the Gamma recurrence $\Gamma(x+3/2) = (x+1/2)\Gamma(x+1/2)$, proved by integration by parts — **not** the reflection formula $\Gamma(s)\Gamma(1-s) = \pi/\sin(\pi s)$, which would re-enter Trap 1. We name this explicitly because the reflection formula is the usual hidden circle in Γ-based arguments, and we do not use it.

The dependency graph has no cycles

NODE	STATEMENT	DEPENDS ON
1	π_0 := factorial limit	— definition
2	$W(x) := \int(1-t^2)^x dt$	— definition
3	recurrence (1)	2 (by parts)
4	$n\, W(n)^2 \to \pi_0/4$	3 + factorial form
5	$\sqrt{x}\, W(x) \to L,\ L^2=\pi_0/4$	4 + Gaussian sandwich
6	$W(1/2) = \pi_0/4$	5 + telescoping (6)
7	$\Gamma(1/2) = \sqrt{\pi_0}$	6 + Beta form
8	closed form for $W(x)$	7
9	$C = (\ln W)''(0)$	8 (also 2, as check)
10	$\zeta(2) = (4-C)/2$	9 (trigamma algebra)
11	$\zeta(2) = \pi_0^2/6$	10 + 6 via shared form

RIGOROUS Every edge points from a lower to a higher node: the graph is acyclic. In particular node 6 (the keystone) does *not* depend on node 11 — breaking the loop the old $O = 2W(1/2)^2$ step had created. The interlock is dissolved, not papered over.

PART VIII · WHY ANY OF THIS WORKS

Periods and motives: the six are one object

Why should six unrelated-looking definitions agree at all? Because each computes a *period* — an integral of an algebraic quantity over an algebraic region — and all six periods belong to the same underlying object.

PLAIN Picture each definition photographing the same statue from a different angle. The photos look different, but they are pictures of one statue. The "statue" is a single algebraic object; π is the measurement attached to it, seen from combinatorics, geometry, or number theory in turn.

RIGOROUS In Grothendieck's language, π is the period of the motive $H^1(G_m)$, the first cohomology of the multiplicative group. The period conjecture predicts all algebraic relations among periods come from morphisms of motives; our trap-free bridges are explicit such morphisms between geometric representatives of this one motive, while the traps mark where a proof would otherwise pick the representative S^1 — the unit circle. The circle is therefore never logically eliminable, only avoidable as an explicit step. This also bounds the program: because all six share one motive, substitutions among them yield tautologies and cannot decide open transcendence questions (e.g. whether π^π is irrational).

PART IX · COMPUTING EACH Π, FAST

Efficient evaluation — and a trap-free accelerator

All six define the same number, but they do not all *compute* it at the same speed. The difference is a property of the estimator, not of the definition: a series pays one term at a time, while an integral can be hit with high-order quadrature. Here is the honest comparison, plus the fastest circle-free route.

VARIABLES

`error ∝ 1/n^p`	convergence order: larger p means each unit of work buys more digits. p = 1 is slow; p = 2 is fast.
`n, m, N, M`	terms, strips, grid size, density bound — the "work" each method spends.
`Richardson extrapolation`	combining two estimates at work n and 2n to cancel the leading error term — a standard, circle-free accelerator.

DEFINITION / ESTIMATOR	NATIVE ORDER	ERROR AT 10^4 WORK	CHARACTER
I Factorial recurrence	$1/n$	7.9×10^{-5}	smooth, from above
V Wallis product	$1/n$	2.4×10^{-4}	smooth, from below
II Volumetric (midpoint)	$1/m^{1.5}$	3.4×10^{-7}	fast (endpoint-limited)
III Lattice count	$1/N$	$\approx2\times10^{-6}$	oscillatory (Gauss circle)
IV Square-free density	$1/\sqrt{M}$	$\approx3\times10^{-5}$	non-monotone
VI Coprime (Monte Carlo)	$1/\sqrt{(\text{samples})}$	noisy	statistical

PLAIN The volume method is fastest in its plain form because a Riemann sum samples the whole curve at once, while the factorial and Wallis series correct themselves only one term at a time. This is about the *method*, not which π is "truer." And every one of them can be sped up without touching a circle.

The trap-free accelerators (all verified to the digits shown)

RIGOROUS — FOUR SPEED-UPS, NONE USING A CIRCLE **(A) Richardson on the volume integral.** Combining midpoint sums at m and 2m, aware of the order-1.5 endpoint, drops the error from 3.4×10^{-7} to $\mathbf{6.4\times10^{-13}}$ at $m = 10^4$ — six extra digits, essentially free.
(B) Gauss–Legendre on the same integral. Reaches the midpoint rule's 10^4-strip accuracy in about **20 nodes**.
(C) Algebraic smoothstep substitution $t = 3w^2 - 2w^3$ (derivative vanishes at both ends, taming the √ singularity; purely algebraic, *no* trig): error $\propto 1/m^2$, e.g. 1.0×10^{-8} at $m = 10^4$.
(D) Richardson on the factorial series. Cancelling the leading $1/n$ term takes the factorial from 7.9×10^{-5} to $\mathbf{4.9\times10^{-10}}$ at $n = 10^4$.

WORKED EXAMPLE — RICHARDSON IN ONE LINE Midpoint sums give $I(m)$ and $I(2m)$ for $4\int\sqrt{(1-t^2)}$. The combination $[2^{1.5}I(2m) - I(m)]/(2^{1.5}-1)$ cancels the leading error. At $m = 1000$ the raw error 1.1×10^{-5} becomes 2.0×10^{-10} — five digits gained from two cheap sums.

NOTE ON THE SLOW Z(2) SERIES The double-series form of C in (10) converges only like $1/K$ when truncated at $k,j \le K$ — honestly slow. But C is also computed by direct quadrature of the clean integrals (9)–(10), which lands on $\zeta(2) = \pi^2/6$ to nine digits at once. The *method* is exact; only one particular series representation of it is slow.

Practical upshot. The volume definition is not only the cleanest conceptually — no circle in sight — it is also the most efficient in native form, and with Richardson or a smoothstep substitution it reaches machine precision in a few thousand operations, entirely circle-free.

PART X · THE AUDIT

Every bridge, against every trap

BRIDGE / RESULT	METHOD	TRAP RISK	STATUS
I ↔ V (Prop 1)	Wallis product	—	Clean
II ↔ V (Eq 4)	rectangular exhaustion	Trap 3	Clean
$W(1/2)$ (Thm 2)	uniqueness + identity (6)	Traps 1,3	Clean
III ↔ V (Thm 3)	Riemann sums	Trap 1	Clean
arcsin (Thm 4)	backward recurrence	Trap 3	Clean
$\Gamma(1/2)$, Gaussian (Thm 5–6)	Beta form + keystone	Trap 1	Clean
$\zeta(2)$ (Thm 7)	$(\ln W)''(0)$ + trigamma	Traps 2,3	Clean
IV ↔ VI	both = $6/\pi^2$	—	Trivial

NUMERICAL VERIFICATION (ALL TO ≥8 DIGITS) Factorial $a_{5000} = 3.14174974$; $W(1/2) = 0.78539816 = \pi/4$; $4W(1/2) = 3.14159265$; lattice $(N=1000) = 3.141549$; arcsin integral $= 1.57079633 = \pi/2$; $\Gamma(1/2) = 1.77245385 = \sqrt{\pi}$; $W'(0) = -0.61370564 = 2\ln 2-2$; $C = 0.71013187 = 4-2\zeta(2)$; $(4-C)/2 = 1.64493407 = \pi^2/6$; $6/\pi^2 = 0.60792710$.

PART XI · HONEST REACH

Where this is useful, and where it is not

The result changes how we *prove* facts about π, not how we *use* π. Stated plainly, the value is concentrated in foundations, special-function theory, computing constants, and teaching — and is essentially nil in the empirical sciences, by the nature of the result.

DOMAIN	BEST USE	RATING
Foundations / reverse math	machine-checkable stratification of proofs by trap content	**Direct**
Special functions	asymptotic Bohr–Mollerup characterization of W	**Direct**
Period theory / motives	elementary, teachable model of the period conjecture	Structural
Computing constants	integer-only, circle-free generation; Richardson-accelerated quadrature	**Direct (niche)**
Pedagogy	dual-register proof of where the circle hides	**Direct**
Engineering / chemistry / biology	none — π is consumed as a constant	Nil
Astrophysics	none	Nil

PLAIN Nobody computing a bridge, a molecule, or an orbit needs to know *which* proof links the faces of π; they use the number itself. The honest worth of this work is to mathematicians who care about the logical structure of constants, and to teachers who want to show, concretely, where the circle hides — and how to walk around it.

APPENDIX · EVERY SYMBOL, ONCE MORE

Master glossary

VARIABLES

π_0	the common value of all six definitions, ≈ 3.14159; = 4W(1/2).
κ	volumetric constant from cone vs. tetrahedron; = $abc/(2r^2h) = \pi_0$.
n!	factorial, product 1·2⋯n. Example 4! = 24.
W(x)	Wallis integral $\int_0^1(1-t^2)^x dt$. W(1) = 2/3.
P(n)	half-step product ∏(2k+1)/(2k+2). P(1) = 3/4.
L	limit $\int_0^R e^{-u^2} du$ as R→∞; = $\sqrt{\pi_0}/2$.
ζ(2)	sum of reciprocal squares 1+¼+1/9+…; = $\pi_0^2/6$.
Γ(x)	Gamma function; Γ(1/2) = $\sqrt{\pi_0}$.
ψ′(s)	trigamma, $\sum 1/(n+s)^2$; ψ′(1) = ζ(2).
B(p,q)	Beta function Γ(p)Γ(q)/Γ(p+q); W(x) = ½B(½,x+1).
C	second log-derivative (ln W)″(0); = 4 − 2ζ(2).
O	odd reciprocal squares $\sum 1/(2k+1)^2$; = $\pi_0^2/8$.
p	a prime number, in products over primes.
$\mathbb{Z}$, $\mathbb{Z}^2$	the integers; the integer grid in the plane.
G_m	the multiplicative group, motivic source of π.

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Provenance, honestly stated

ACKNOWLEDGMENTS The derivations, figures, code, and prose of this account were developed with substantial assistance from an AI system (Anthropic's Claude), used as a working collaborator for symbolic derivation, adversarial proof-checking, numerical verification, and drafting. All mathematical claims were independently verified by the author, symbolically and numerically (8–40 significant digits). Responsibility for the work, and for any remaining error, rests with the author. A key circularity in the ζ(2) bridge (Part VII) was first surfaced by adversarial review through a second AI system, and is gratefully recorded as the kind of external check that strengthens a result.

ON NOVELTY The classical ingredients are old and are attributed throughout: Wallis's product (1656), the Bohr–Mollerup characterization (1922), Euler's product and the Basel value (1737), Gaussian lattice counting (Gauss, 1801), and Hermite-era special-function identities. **The contribution claimed here is the proof architecture**: the trap taxonomy, the no-evaluation strategy, the telescoping keystone, and the trap-free ζ(2) closure that assembles all six definitions into one acyclic dependency graph. This is offered as a foundational/expository advance, not as a new theorem about the value of π.

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A Circle-Free Theory of π · the complete account · S. B. Anderson, June 2026. Unifies the keystone (Rev. 4), the ζ(2) closure (Rev. 5), the interlock repair (Rev. 6), the six-definition bridges, and the utilities survey. Verified symbolically and numerically; see Acknowledgments for provenance and the AI-assistance statement.